

Statistical Analysis I

(C4)

03

Methods and Studies with Users

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1

Overview

Statistical Analysis

Preparing data for statistical analysis

Descriptive statistics

Types of data

2

01

PREPARING DATA FOR STATISTICAL ANALYSIS

3

Introduction

Original data collected from

lab-based experiments, usability tests, field studies,
surveys, etc.

**Need to be carefully processed before any
statistical analysis can be conducted**

4

Reasons for the need for preprocessing

Cleaning up data

- Detect errors
- Formatting

Coding

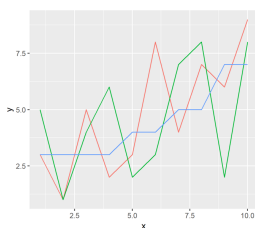
- Types of data that need to be coded (e.g. gender)
- Be consistent (e.g. more than one person)

Organizing the data

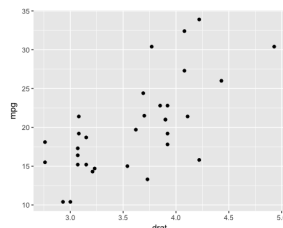
- Accommodate to the requirements of statistical software

5

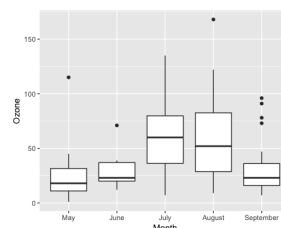
Before Analyzing Visualize



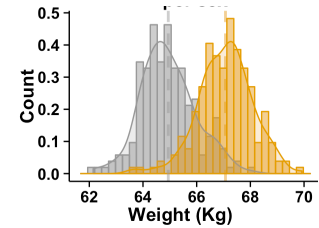
Line Graph



Scatter Plot



Box Plot



Histogram

6

Importance of Visualizing

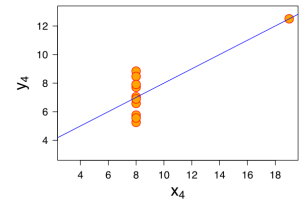
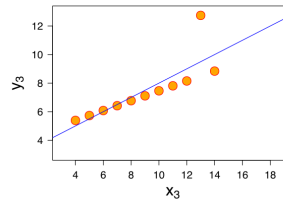
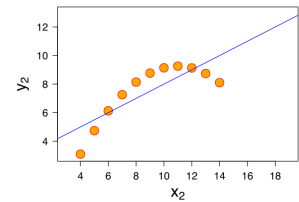
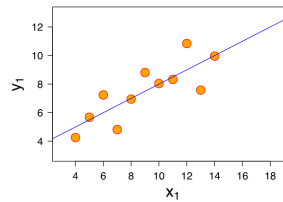
Anscombe's quartet

Four datasets (two variables each)

Identical simple descriptive statistics (table)

Very different distributions (figure)

Sample property	Value
Mean $\bar{x}$	9
Variance of x	11
Mean $\bar{y}$	7.50 (to 2 decimal places)
Variance of y	4.122 or 4.127 (to 3 decimal places)
Correlation between x and y	0.816 (to 3 decimal places)
Linear regression line	$y = 3.00 + 0.500x$ (to 2 and 3 decimal places, respectively)



7

02

DESCRIPTIVE STATISTICS

8

Descriptive statistics

Measures of central tendency

Mean

Arithmetic average

Median

Middle value

Mode

Value that occurs with the greatest frequency

9

Descriptive statistics

Measures of spread

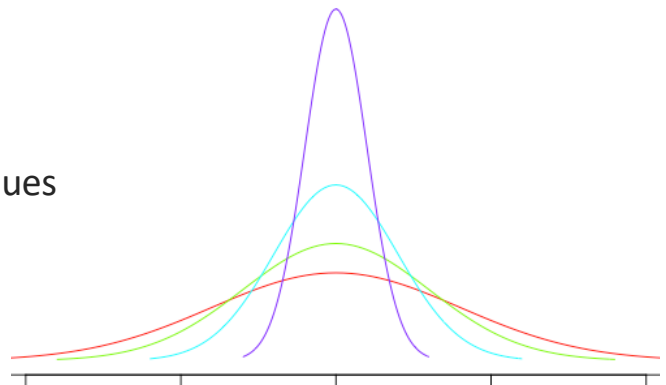
Range

Distance between
the highest and lowest values

Variance

Standard deviations

Interquartile range (IQR)



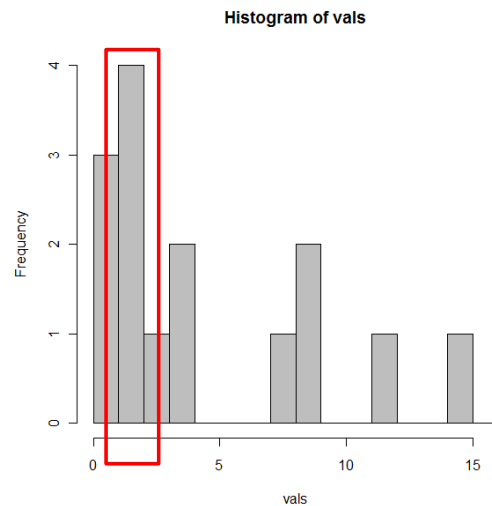
Higher ranges, variances or standard deviations
indicate that the data set is more distributed

10

Mode - Example

1, 1, 1, 2, 2, 2, 2, 3,
4, 4, 8, 9, 9, 12, 15

= 2



11

Median vs. Mean -> Outliers

9, 4, 15, 2, 8, 1, 1, 4, 2, 2, 12, 3, 2, 1, 9

Sort

1, 1, 1, 2, 2, 2, 2, 3, 4, 4, 8, 9, 9, 12, 15

Median = 3 / Mean = 5

-> Some values very big -> Outliers

12

Outliers

3, 2, 4, 3, 1, 4, 3, 2, 2, 3, 3, 20, 1, 2, 3, 2, 2, 3, 4

13

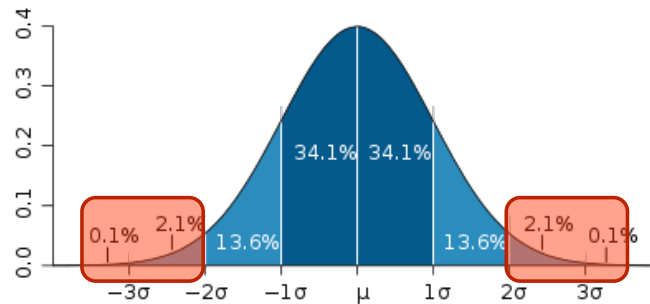
Outliers

3, 2, 4, 3, 1, 4, 3, 2, 2, 3, 3, 20, 1, 2, 3, 2, 2, 3, 4

14

Outliers – How to Detect them?

Several ways to compute, but ...

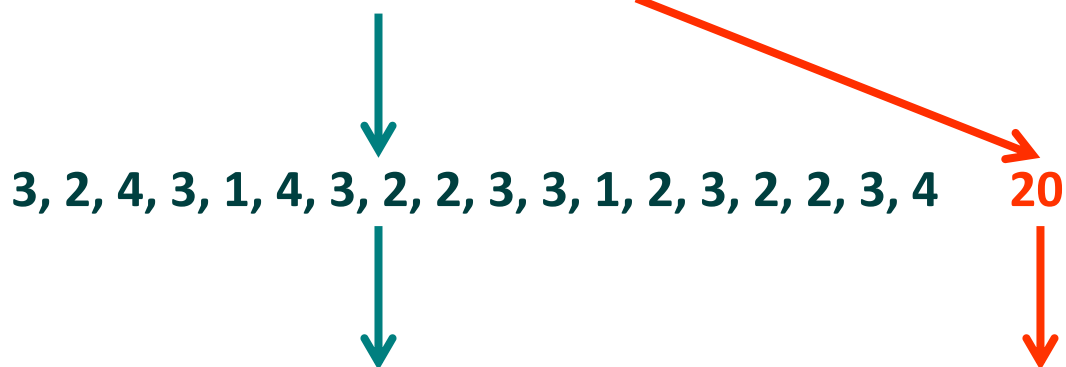


Out of $[\bar{x} - 2s, \bar{x} + 2s]$

15

... But, don't ignore them!

3, 2, 4, 3, 1, 4, 3, 2, 2, 3, 3, **20**, 1, 2, 3, 2, 2, 3, 4



mean, mode, median, stdev, etc.

Analyse one by one

16

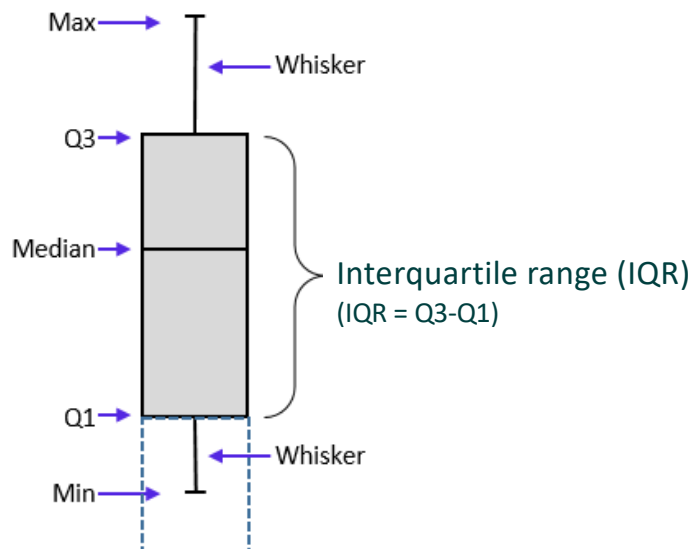
Quartiles / Percentiles

Left **25%**, **50%**, **75%** of the values before

1, 1, 1, **2**, 2, 2, 2, **3**, 4, 4, 8, **9**, 9, 12, 15

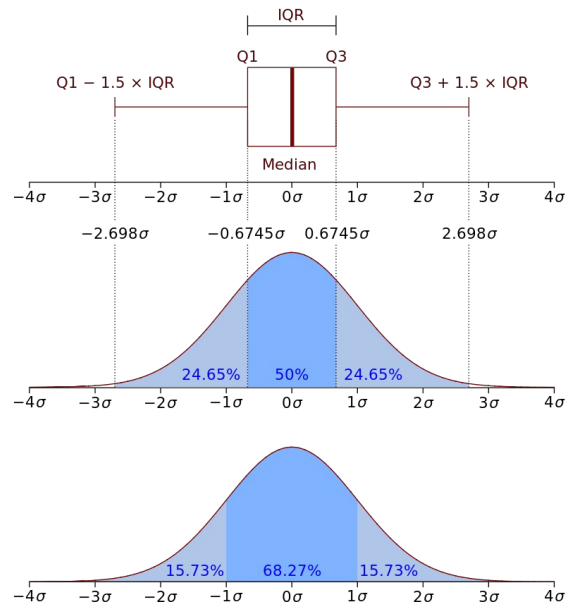
17

Boxplot



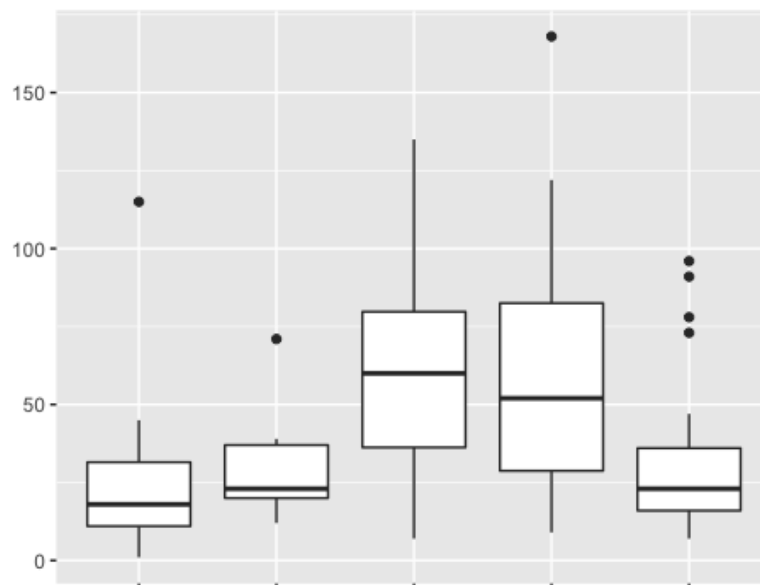
18

Boxplot



19

Boxplot – Comparing several conditions



20

03

TYPES OF DATA

21

Types of Data

Nominal

Indicates categories; no notion of ordering
e.g., Techniques, Gender, Occupation

Ordinal

Data can be order; difference between any of two values not always equal
e.g., Likert scales

Interval

Data can be order; difference between any of two values are the same; there is no absolute zero
e.g., temperature

Ratio

Data can be order; difference between any of two values are the same; there is the absolute zero
e.g., weight, height, length, time, speed, error rate, counts

22

Types of Data

Levels of Measurement

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

23

What measures and When?

Nominal: mode

Eg., "How many prefer A, B, or C"

Ordinal: median, mode, quartiles

Eg., Likert scales - "Level of satisfaction, 1 to 5"

Interval / Ratio: mean, median, quartiles, StDv

Eg., "Average time to complete a task"

24

Summary

Statistical Analysis

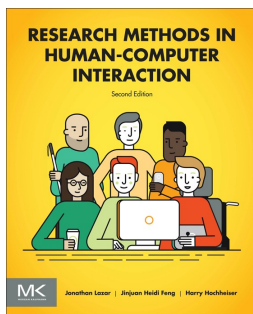
Preparing data for statistical analysis

Descriptive statistics

Types of data

25

Credits



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26